

1. First Non-Repeating Character

- Find the first non-repeating character in a string using HashMap.

Code:

```
import java.util.*;

public class FirstNonRepeating {

    public static Character findFirstNonRepeating(String str) {

        Map<Character, Integer> map = new HashMap<>();

        for (char ch : str.toCharArray()) {

            map.put(ch, map.getOrDefault(ch, 0) + 1);

        }

        for (char ch : str.toCharArray()) {

            if (map.get(ch) == 1) {

                return ch;

            }

        }

        return null;

    }

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter the string: ");

        String input = sc.next();

        System.out.print("The first non repeating character: ");

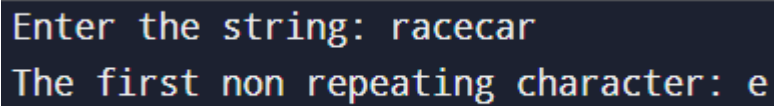
        System.out.println(findFirstNonRepeating(input));

    }

}
```

```
}  
}
```

Output:



```
Enter the string: racecar  
The first non repeating character: e
```

2. Online Shopping Cart

- Use List for cart, Map for product-price mapping.

Code:

```
import java.util.*;
```

```
public class ShoppingCart {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        Map<String, Double> productPrice = new HashMap<>();
```

```
        System.out.print("Enter number of products: ");
```

```
        int n = sc.nextInt();
```

```
        sc.nextLine();
```

```
        for (int i = 0; i < n; i++) {
```

```
            System.out.print("Enter product name: ");
```

```
            String name = sc.nextLine();
```

```
            System.out.print("Enter price: ");
```

```
            double price = sc.nextDouble();
```

```

        sc.nextLine();

        productPrice.put(name, price);
    }

    List<String> cart = new ArrayList<>();

    System.out.print("Enter number of items to add to cart: ");
    int m = sc.nextInt();
    sc.nextLine();

    for (int i = 0; i < m; i++) {
        System.out.print("Enter " + (i+1) + "st product to add: ");
        String item = sc.nextLine();

        if (productPrice.containsKey(item)) {
            cart.add(item);
        } else {
            System.out.println("Product not found!");
        }
    }

    double total = 0;
    System.out.println("\nCart items:");
    for (String item : cart) {
        System.out.println(item + " - " + productPrice.get(item));
        total += productPrice.get(item);
    }

```

```
        System.out.println("Total price: " + total);

        sc.close();
    }
}
```

Output:

```
Enter number of products: 3
Enter product name: Mobile
Enter price: 5000
Enter product name: Headphones
Enter price: 2000
Enter product name: Charger
Enter price: 500
Enter number of items to add to cart: 2
Enter 1st product to add: Mobile
Enter 2st product to add: Charger

Cart items:
Mobile - 5000.0
Charger - 500.0
Total price: 5500.0
```

3. Library Management

- Use Map to track availability.

Code:

```
import java.util.*;
```

```
class Book {
```

```
    String title;
```

```
    String author;
```

```
    public Book(String title, String author) {
```

```

        this.title = title;

        this.author = author;
    }

    public boolean equals(Object obj) {
        Book other = (Book) obj;

        return title.equals(other.title) && author.equals(other.author);
    }

    public int hashCode() {
        return title.hashCode() + author.hashCode();
    }

    public String toString() {
        return title + " by " + author;
    }

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        Map<Book, Boolean> library = new HashMap<>();

        System.out.print("Enter number of books: ");
        int n = sc.nextInt();
        sc.nextLine();

        for (int i = 0; i < n; i++) {
            System.out.print("Enter title: ");

            String title = sc.nextLine();

```

```

        System.out.print("Enter author: ");
        String author = sc.nextLine();

        library.put(new Book(title, author), true);
    }

    System.out.print("\nEnter title to issue: ");
    String issueTitle = sc.nextLine();

    System.out.print("Enter author: ");
    String issueAuthor = sc.nextLine();

    Book issueBook = new Book(issueTitle, issueAuthor);

    if (library.containsKey(issueBook) && library.get(issueBook)) {
        library.put(issueBook, false);
        System.out.println("Book issued successfully.");
    } else {
        System.out.println("Book not available.");
    }

    System.out.println("\nLibrary status:");
    for (Map.Entry<Book, Boolean> entry : library.entrySet()) {
        System.out.println(entry.getKey() + " -> " +
            (entry.getValue() ? "Available" : "Issued"));
    }

    sc.close();

```

```
}  
}
```

Output:

```
Enter number of books: 2  
Enter title: Ikigai  
Enter author: Francesc Miralles  
Enter title: Atomic Habits  
Enter author: James Clear  
  
Enter title to issue: Atomic Habits  
Enter author: James Clear  
Book issued successfully.  
  
Library status:  
Atomic Habits by James Clear -> Issued  
Ikigai by Francesc Miralles -> Available
```